



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab Programs

SET 13- Geographic Data

13A. Map Chart

Program:

```
install.packages("ggplot2")
install.packages("plotly")
install.packages("knitr")

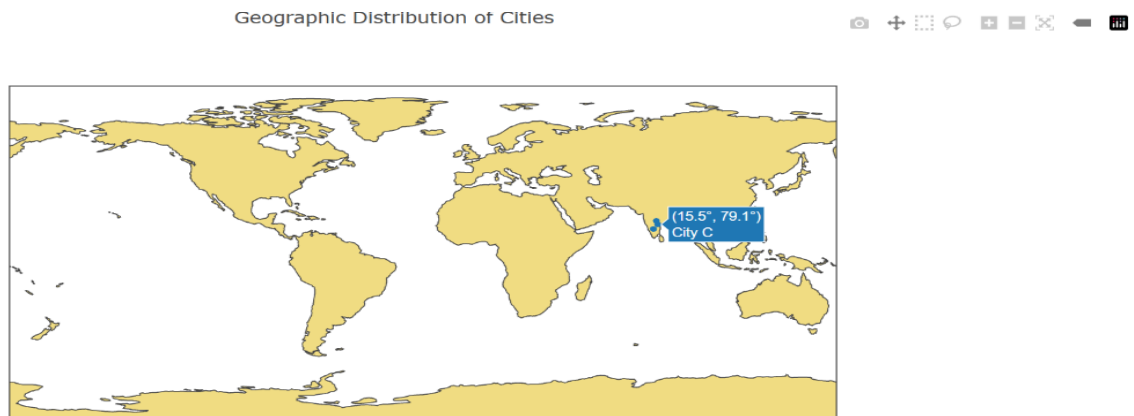
library(ggplot2)
library(plotly)
library(knitr)

geo_data <- data.frame(
  City=c("City A","City B","City C"),
  Population=c(500000,700000,600000),
  AvgTemperature=c(75,68,80),
  Elevation=c(1000,800,1200),
  Longitude=c(78.5,77.2,79.1),
  Latitude=c(17.4,13.0,15.5))

plot_ly(
  geo_data,
  type="scattergeo",
  lon=~Longitude,
  lat=~Latitude,
  text=~City,
  mode="markers"
) %>%
layout(
  title="Geographic Distribution of Cities",
```

```
geo=list(showland=TRUE))
```

Output:

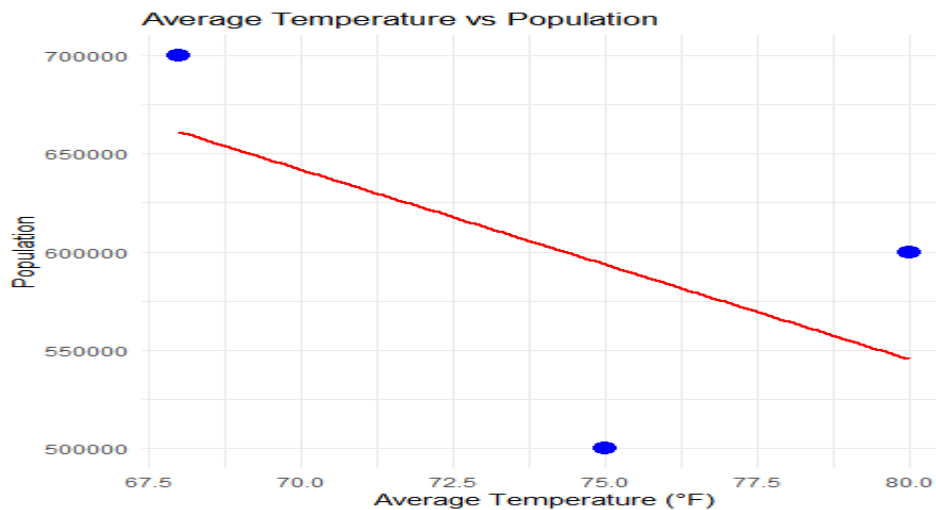


13B. Scatter Plot

Program:

```
ggplot(geo_data,  
aes(x=AvgTemperature,  
y=Population))+  
geom_point(size=4,color="blue")+  
geom_smooth(method="lm",se=FALSE,color="red")+  
labs(title="Average Temperature vs Population",  
x="Average Temperature (°F)",  
y="Population")+  
theme_minimal()
```

Output:



13C. Table

Program:

```
kable(  
  geo_data,  
  caption="Geographic Data of Cities"  
)
```

Output:

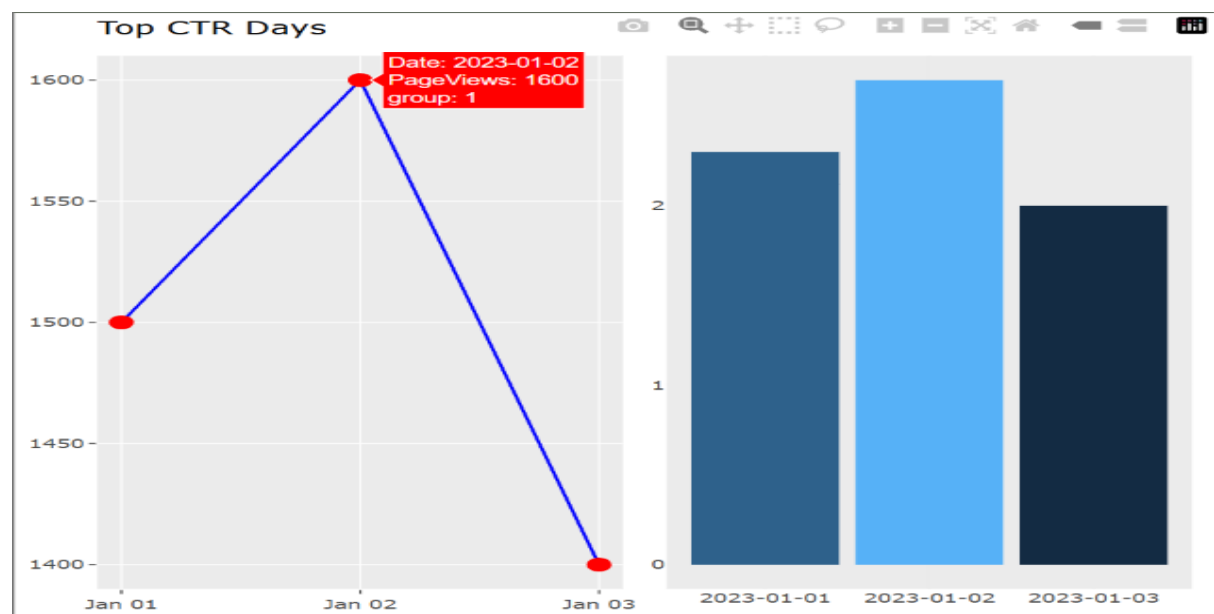
Table: Geographic Data of Cities					
City	Population	AvgTemperature	Elevation	Longitude	Latitude
City A	5e+05	75	1000	78.5	17.4
City B	7e+05	68	800	77.2	13.0
City C	6e+05	80	1200	79.1	15.5

13D. Interactive Dashboard

Program:

```
htmlwidgets::saveWidget(  
  dashboard,  
  "GeographicDashboard.html",  
  selfcontained=TRUE)  
browseURL("GeographicDashboard.html")
```

Output:



SET 14- Survey Results

14A. Stacked Bar Chart

Program:

```
install.packages("ggplot2")
install.packages("plotly")
install.packages("knitr")
install.packages("fmsb")
install.packages("reshape2")
survey_data <- data.frame(
  Respondent=c(1,2,3),
  Question1=c("A","B","C"),
  Question2=c("B","A","A"),
  Question3=c("C","D","B"))
survey_long <- melt(
  survey_data,
  id.vars="Respondent",
  variable.name="Question",
  value.name="Response")
ggplot(
  survey_long[survey_long$Question=="Question1",],
  aes(x=Question,fill=Response)
)+
  geom_bar()+
  labs(
    title="Distribution of Answers for Question 1",
    x="Question",
    y="Count",
    fill="Response"
  )+
  theme_minimal()
```

Output:

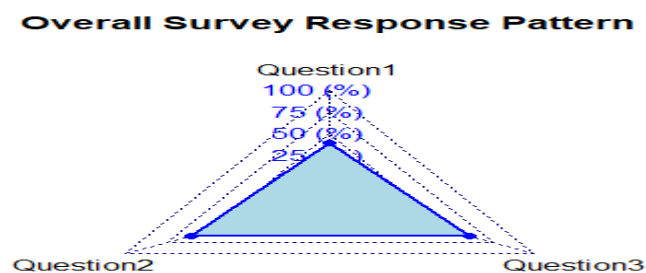


14B. Radar Chart

Program:

```
response_data <- data.frame(  
  Question=c("Question1","Question2","Question3"),  
  Score=c(2,3,3))  
  
radar <- data.frame(  
  Max=c(5,5,5),  
  Min=c(0,0,0),  
  Average=response_data$Score)  
colnames(radar)=c("Question1","Question2","Question3")  
radarchart(radar,axistype=1,  
  pcol="blue",  
  pfc="lightblue",plwd=2,  
  title="Overall Survey Response Pattern")
```

Output:



14C. Table

Program:

```
kable(  
  survey_data,  
  caption="Survey Response Data"  
)
```

Output:

Table: Survey Response Data

Respondent	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

14D. Interactive Dashboard

Program:

```
htmlwidgets::saveWidget(  
  dashboard,  
  "SurveyDashboard.html",  
  selfcontained=TRUE)  
browseURL("SurveyDashboard.html")
```

Output:

Show entries

Search:

Survey Responses

	Respondent	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

Showing 1 to 3 of 3 entries

Previous Next

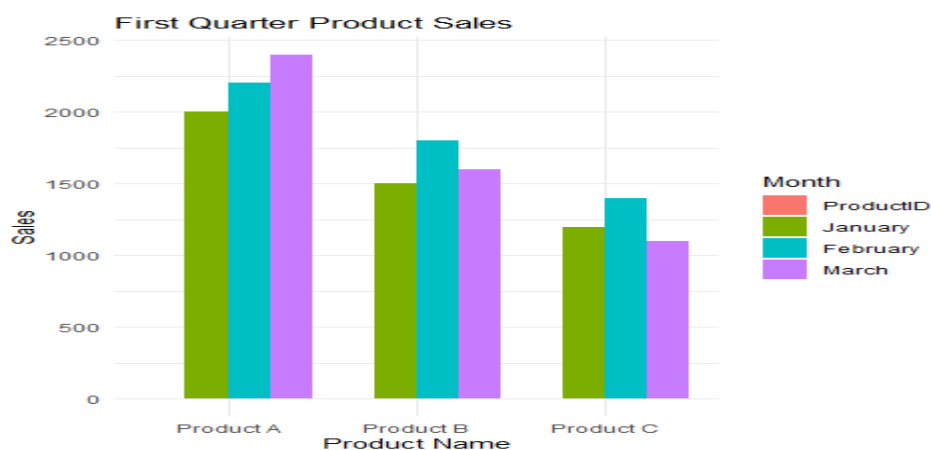
SET 15- Product Sales Analysis

15A. Grouped Bar Chart

Program:

```
sales_data <- data.frame(  
  ProductID=c(1,2,3),  
  ProductName=c("Product A","Product B","Product C"),  
  January=c(2000,1500,1200),  
  February=c(2200,1800,1400),  
  March=c(2400,1600,1100))  
  
sales_long <- melt(sales_data,  
  id.vars="ProductName",  
  variable.name="Month",  
  value.name="Sales")  
  
ggplot(sales_long,  
  aes(x=ProductName,  
  y=Sales,  
  fill=Month))+  
  geom_bar(stat="identity",position="dodge")+  
  labs(title="First Quarter Product Sales",  
  x="Product Name",  
  y="Sales")+  
  theme_minimal()
```

Output:

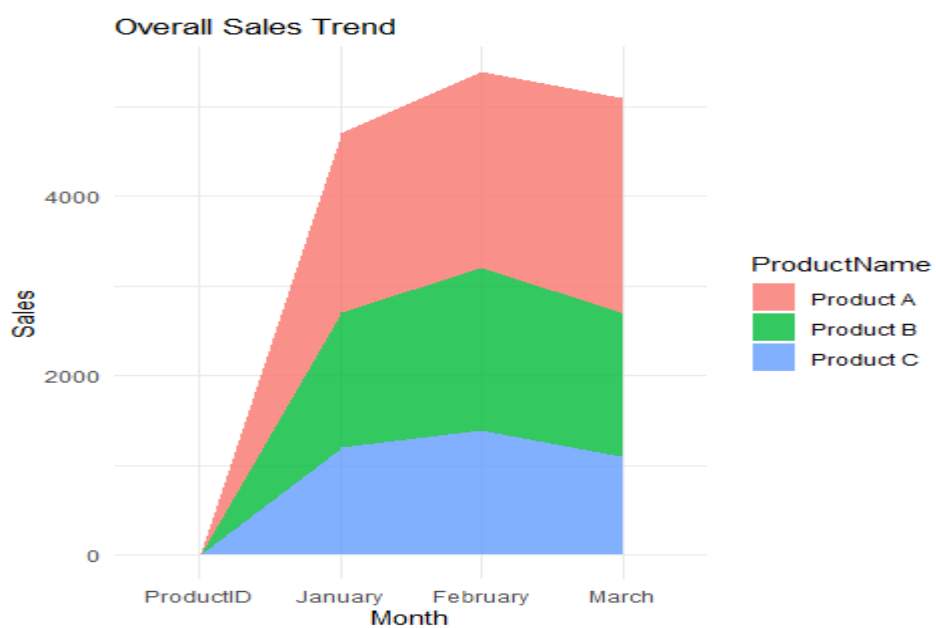


15B. Stacked Area Chart

Program:

```
ggplot(sales_long,  
aes(x=Month,  
y=Sales,  
fill=ProductName,  
group=ProductName))+  
geom_area(alpha=0.8)+  
labs(title="Overall Sales Trend",  
x="Month",  
y="Sales")+  
theme_minimal()
```

Output:



15C. Interactive Table

Program:

```
datatable(  
sales_data,  
caption="Monthly Product Sales",  
options=list(pageLength=10)  
)
```

Output:

Show entries

Search:

Monthly Product Sales

ProductID	ProductID	ProductName	January	February	March
1	1	Product A	2000	2200	2400
2	2	Product B	1500	1800	1600
3	3	Product C	1200	1400	1100

Showing 1 to 3 of 3 entries

Previous Next

15D. 4: Interactive Dashboard

Program:

```
library(plotly)

bar_plot <- ggplotly(
  ggplot(sales_long,
    aes(x=ProductName,
      y=Sales,
      fill=Month)) +
    geom_bar(stat="identity", position="dodge") +
    labs(title="Grouped Bar Chart") +
    theme_minimal()
)

area_plot <- ggplotly(
  ggplot(sales_long,
    aes(x=Month,
      y=Sales,
      fill=ProductName,
      group=ProductName)) +
    geom_area(alpha=0.8) +
    labs(title="Stacked Area Chart") +
    theme_minimal()
)

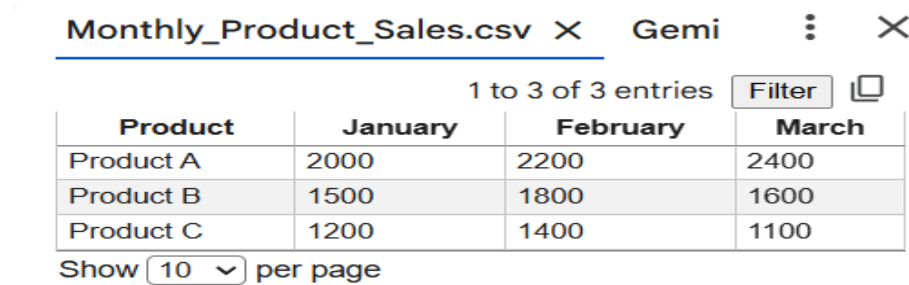
dashboard <- subplot(
  bar_plot,
  area_plot,
  nrows=1,
```

shareX=FALSE,

shareY=FALSE)

dashboard

Output:



Monthly_Product_Sales.csv X Gemi ⋮ X			
1 to 3 of 3 entries Filter			
Product	January	February	March
Product A	2000	2200	2400
Product B	1500	1800	1600
Product C	1200	1400	1100

Show 10 per page

SET 16- Customer Demographics Analysis

16A. Line Chart

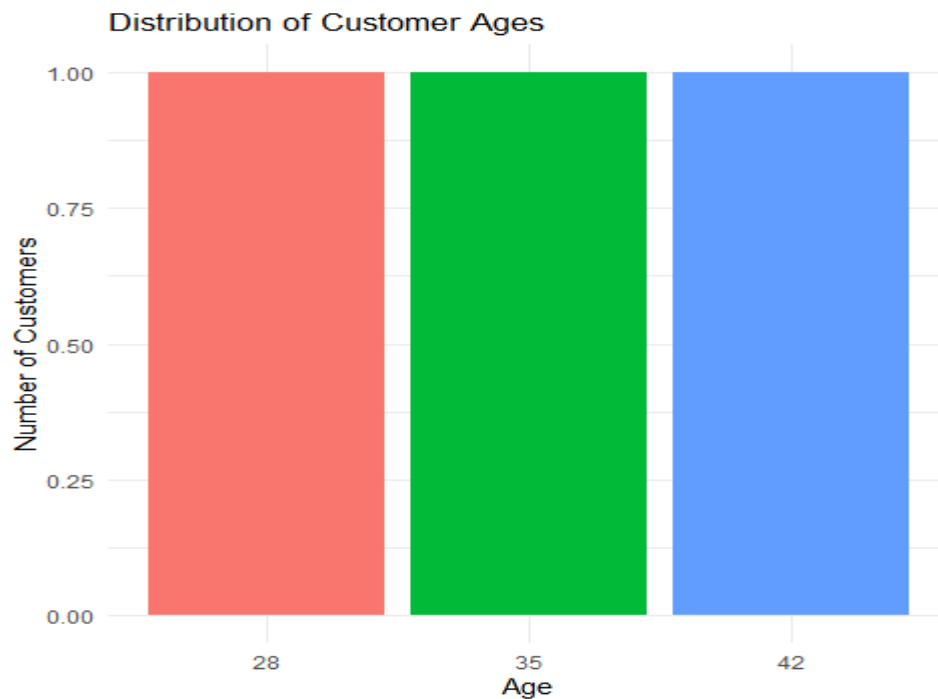
Program:

```
library(ggplot2)

customer_data <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

ggplot(customer_data, aes(x = factor(Age), fill = factor(Age))) +
  geom_bar() +
  labs(
    title = "Distribution of Customer Ages",
    x = "Age",
    y = "Number of Customers"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Output:



16B. Bar Chart

Program:

```
library(ggplot2)

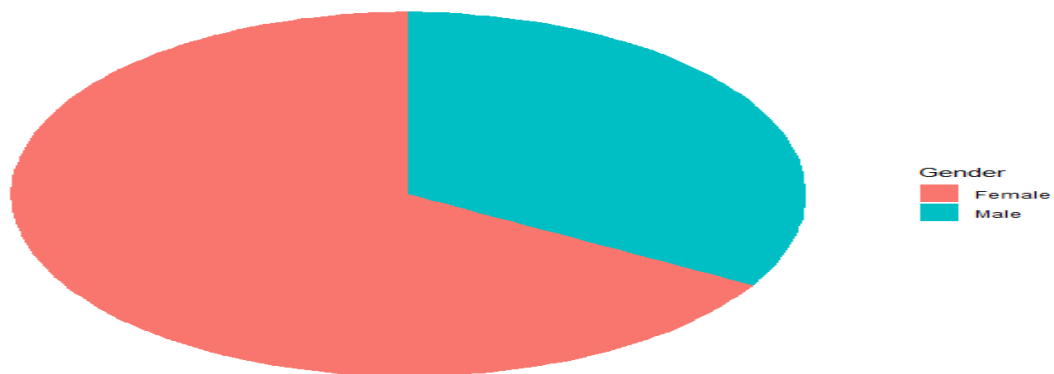
customer_data <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

gender_data <- as.data.frame(table(customer_data$Gender))
colnames(gender_data) <- c("Gender","Count")

ggplot(gender_data, aes(x = "", y = Count, fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Distribution of Customers by Gender") +
  theme_void()
```

Output:

Distribution of Customers by Gender



16C. Table

Program:

```
View(customer_data)
```

Output:

Data: customer_data				
	CustomerID	Age	Gender	Income
1	1	28	Female	50000
2	2	35	Male	60000
3	3	42	Female	75000

16D. Tableau Dashboard

Program:

```
CustomerID, Age, Gender, Income
```

```
1, 28, Female, 50000
```

```
2, 35, Male, 60000
```

```
3, 42, Female, 75000
```

Output:

CustomerID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

SET 17- Employee Performance Analysis

17A. Line Chart

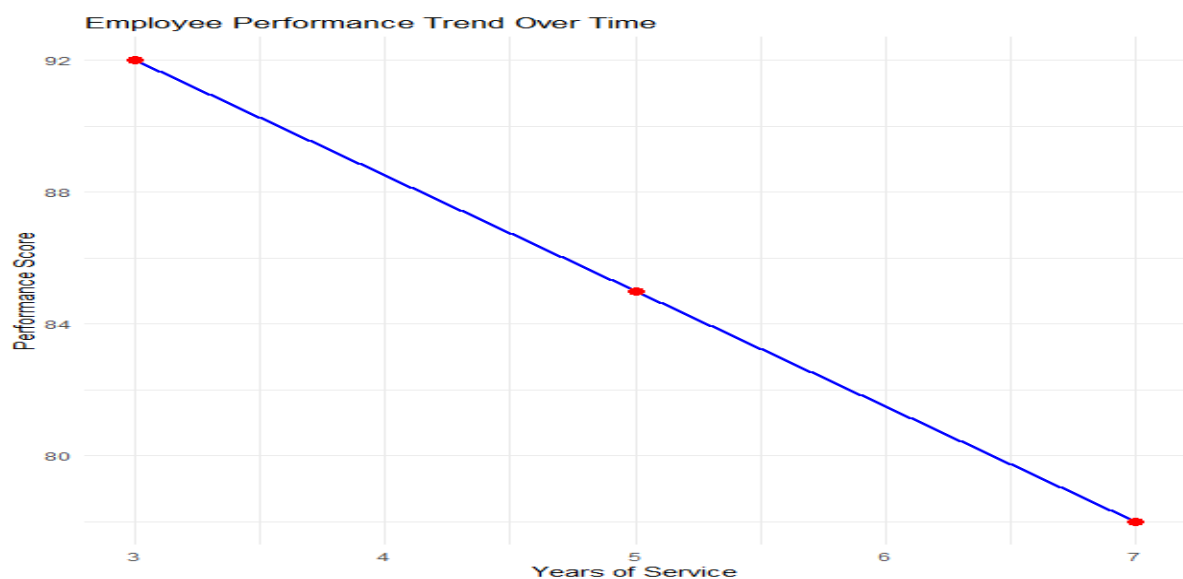
Program:

```
library(ggplot2)

employee_data <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsOfService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)

ggplot(employee_data,
  aes(x = YearsOfService, y = PerformanceScore, group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(size = 3, color = "red") +
  labs(
    title = "Employee Performance Trend Over Time",
    x = "Years of Service",
    y = "Performance Score"
  ) +
  theme_minimal()
```

Output:



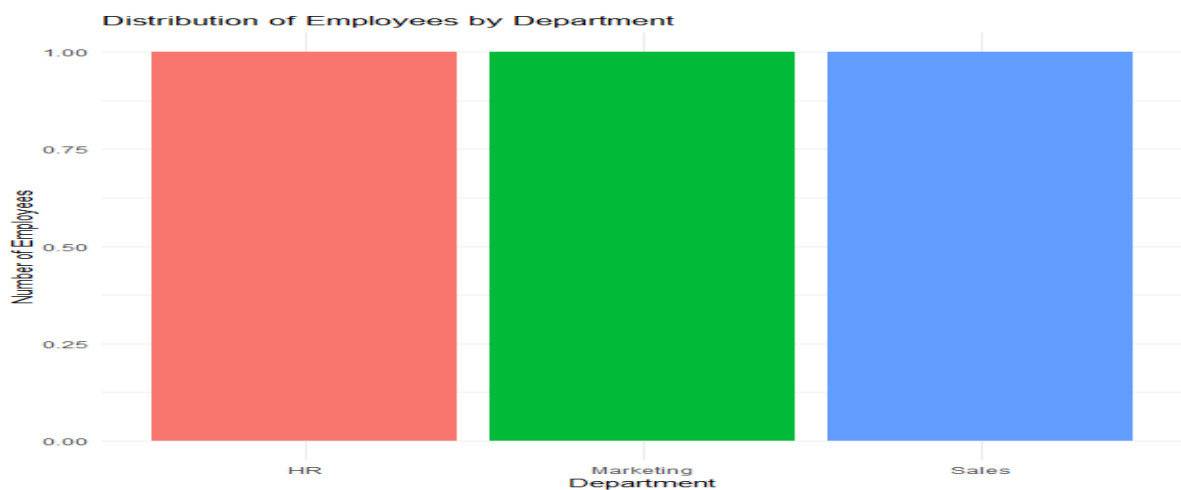
17B. Bar Chart

Program:

```
library(ggplot2)

ggplot(employee_data,
       aes(x = Department, fill = Department)) +
  geom_bar() +
  labs(
    title = "Distribution of Employees by Department",
    x = "Department",
    y = "Number of Employees" ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Output:



17C. Table

Program:

```
View(employee_data)
```

Output:

Data: employee_data				
	EmployeeID	Department	YearsOfService	PerformanceScore
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

17D. Tableau Dashboard

Program:

```
write.csv(employee_data,  
          "Employee_Performance.csv",  
          row.names = FALSE)
```

Output:

EmployeeID	Department	YearsOfService	PerformanceScore
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78